Understandin-Diabetes-Progression.R

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2026-10-22

```
################################################################################
# DATA ANALYTICS AND STATISTICS WITH APPLICATIONS – FINAL PROJECT
# SCENARIO 1: UNDERSTANDING DIABETES PROGRESSION
# Fully Reproducible, Documented, and Commented R Script
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# Date: 2026-10-22
################################################################################

# ==============================================================================
# 0. SETUP AND AUTOMATIC PACKAGE MANAGEMENT
# ==============================================================================

# List of required packages for the data analysis workflow
needed_packages <- c("ggplot2", "dplyr", "broom")

# Automatically check for, install, and load missing dependencies
for (pkg in needed_packages) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg)
  }
}

# Load libraries into the active working environment
library(ggplot2)
```

```
## Warning: il pacchetto 'ggplot2' è stato creato con R versione 4.6.1
```

```
library(dplyr)
```

```
## Warning: il pacchetto 'dplyr' è stato creato con R versione 4.6.1
```

```
##
## Caricamento pacchetto: 'dplyr'
```

```
## I seguenti oggetti sono mascherati da 'package:stats':
##
##     filter, lag
```

```
## I seguenti oggetti sono mascherati da 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(broom)
```

```
## Warning: il pacchetto 'broom' è stato creato con R versione 4.6.1
```

```r
# Optimize print formatting for numeric summaries (4 significant digits)
options(digits = 4)

# ============================================================================
# QUESTION 1. UNDERSTAND AND PREPARE THE DATA
# ============================================================================

# Define the source data file name (must reside in the same working directory)
data_file <- "diabetes.tab.txt"

# Dynamic path error handling: fallback to manual file selection dialog
if (!file.exists(data_file)) {
  message("WARNING: 'diabetes.tab.txt' was not found in the current working directory.")
  message("Please select the supplied data file manually using the dialog box.")
  data_file <- file.choose()
}
```

```
## WARNING: 'diabetes.tab.txt' was not found in the current working directory.
```

```
## Please select the supplied data file manually using the dialog box.
```

```r
# Import the official tab-separated dataset (NCSU Source)
diabetes <- read.table(data_file, header = TRUE, sep = "\t")

# Convert column names to lowercase for typing consistency and fewer syntax errors
names(diabetes) <- tolower(names(diabetes))

# Rename the outcome variable 'y' to 'progression' for clinical clarity
if ("y" %in% names(diabetes)) {
  diabetes <- diabetes %>% rename(progression = y)
}

# Preliminary structure and dimension audit of the data frame
dim(diabetes)  # Expected output: 442 observations and 11 variables
```

```
## [1] 442  11
```

```r
names(diabetes)
```

```
##  [1] "age"         "sex"         "bmi"         "bp"          "s1"
##  [6] "s2"          "s3"          "s4"          "s5"          "s6"
## [11] "progression"
```

```r
str(diabetes)
```

```
## 'data.frame':    442 obs. of  11 variables:
##  $ age        : num  59 48 72 24 50 23 36 66 60 29 ...
##  $ sex        : num  2 1 2 1 1 1 2 2 2 1 ...
##  $ bmi        : num  32.1 21.6 30.5 25.3 23 22.6 22 26.2 32.1 30 ...
##  $ bp         : num  101 87 93 84 101 89 90 114 83 85 ...
##  $ s1         : num  157 183 156 198 192 139 160 255 179 180 ...
##  $ s2         : num  93.2 103.2 93.6 131.4 125.4 ...
##  $ s3         : num  38 70 41 40 52 61 50 56 42 43 ...
##  $ s4         : num  4 3 4 5 4 2 3 4.55 4 4 ...
##  $ s5         : num  4.86 3.89 4.67 4.89 4.29 ...
##  $ s6         : num  87 69 85 89 80 68 82 92 94 88 ...
##  $ progression: num  151 75 141 206 135 97 138 63 110 310 ...
```

```
# Explicitly encode the 'sex' variable as a factor (categorical attribute)
diabetes$sex <- factor(diabetes$sex)

# Data Quality Audit: Check for missing data cells (NAs) or duplicate records
missing_summary <- colSums(is.na(diabetes))
print("--- Missing Values Audit Summary ---")
```

```
## [1] "--- Missing Values Audit Summary ---"
```

```
print(missing_summary) # Should show 0 for all columns
```

```
##         age         sex         bmi          bp          s1          s2
##           0           0           0           0           0           0
##          s3          s4          s5          s6 progression
##           0           0           0           0           0
```

```
total_duplicates <- sum(duplicated(diabetes))
print(paste("Duplicate records detected:", total_duplicates)) # Should equal 0
```

```
## [1] "Duplicate records detected: 0"
```

```
# ============================================================================
# QUESTION 2. EXPLORE THE DATA
# ============================================================================

# Define a helper function to calculate comprehensive core numeric summaries
describe_numeric <- function(x) {
  c(
    n = sum(!is.na(x)),
    mean = mean(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    sd = sd(x, na.rm = TRUE),
    min = min(x, na.rm = TRUE),
    q1 = as.numeric(quantile(x, 0.25, na.rm = TRUE)),
    q3 = as.numeric(quantile(x, 0.75, na.rm = TRUE)),
    max = max(x, na.rm = TRUE),
    iqr = IQR(x, na.rm = TRUE)
  )
}

# Compute baseline descriptive metrics for the outcome and key predictors
print("--- Disease Progression Metrics ---")
```

```
## [1] "--- Disease Progression Metrics ---"
```

```
print(describe_numeric(diabetes$progression))
```

```
##      n   mean median     sd    min     q1     q3    max    iqr
## 442.00 152.13 140.50  77.09  25.00  87.00 211.50 346.00 124.50
```

```
print("--- Body Mass Index (BMI) Metrics ---")
```

```
## [1] "--- Body Mass Index (BMI) Metrics ---"
```

```
print(describe_numeric(diabetes$bmi))
```

```
##       n    mean  median      sd     min      q1      q3     max     iqr
## 442.000  26.376  25.700   4.418  18.000  23.200  29.275  42.200   6.075
```

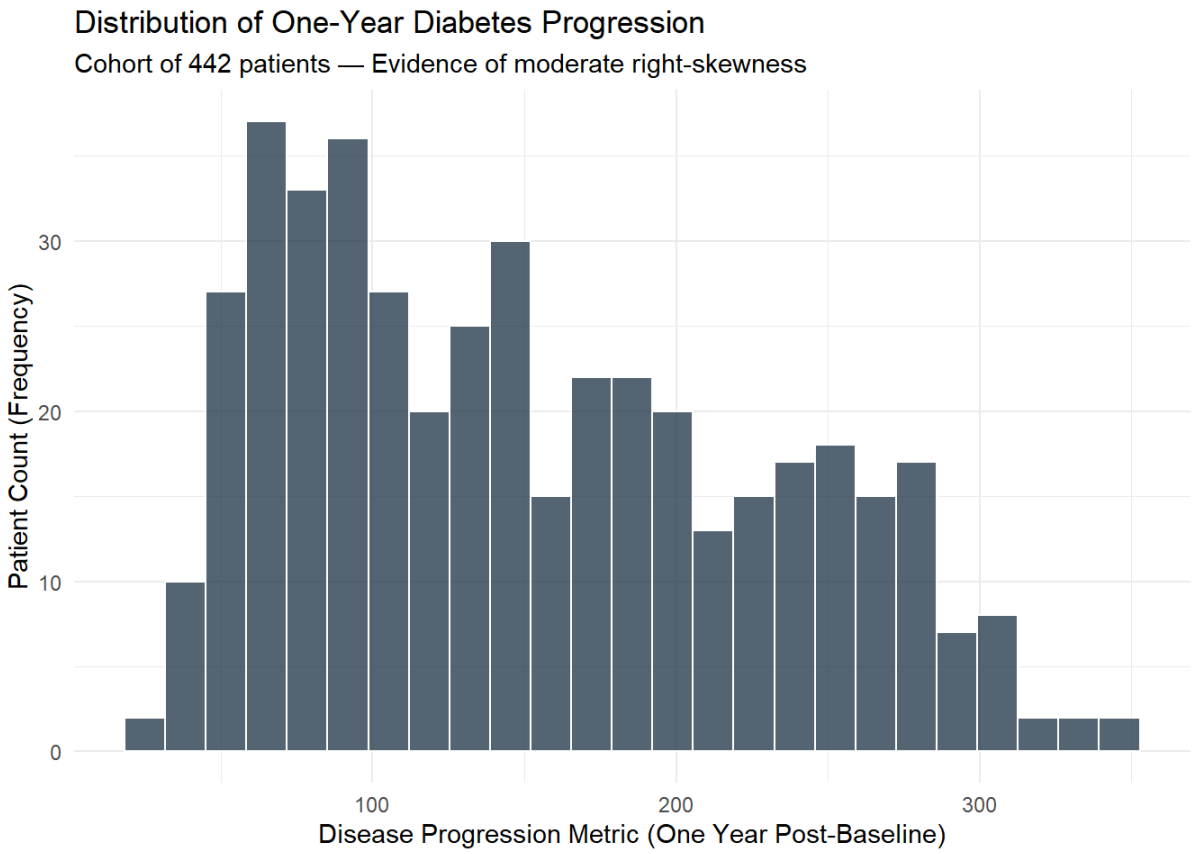
```
print("--- Patient Age Distribution Metrics ---")
```

```
## [1] "--- Patient Age Distribution Metrics ---"
```

```
print(describe_numeric(diabetes$age))
```

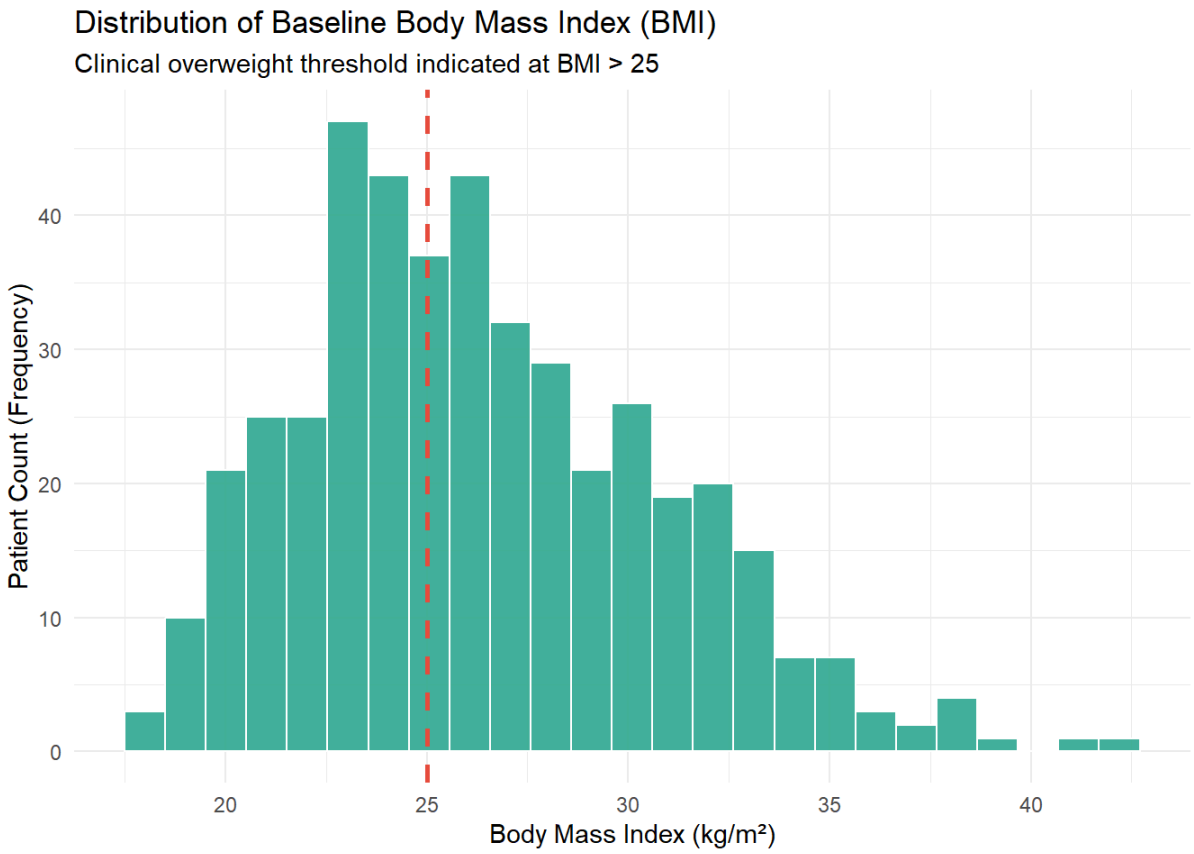
```
##      n   mean median     sd    min     q1     q3    max    iqr
## 442.00  48.52  50.00  13.11  19.00  38.25  59.00  79.00  20.75
```

```
# Visualization 1: Histogram mapping diabetes progression distribution
ggplot(diabetes, aes(x = progression)) +
  geom_histogram(bins = 25, fill = "#2c3e50", colour = "white", alpha = 0.8) +
  labs(
    title = "Distribution of One-Year Diabetes Progression",
    subtitle = "Cohort of 442 patients — Evidence of moderate right-skewness",
    x = "Disease Progression Metric (One Year Post-Baseline)",
    y = "Patient Count (Frequency)"
  ) +
  theme_minimal()
```

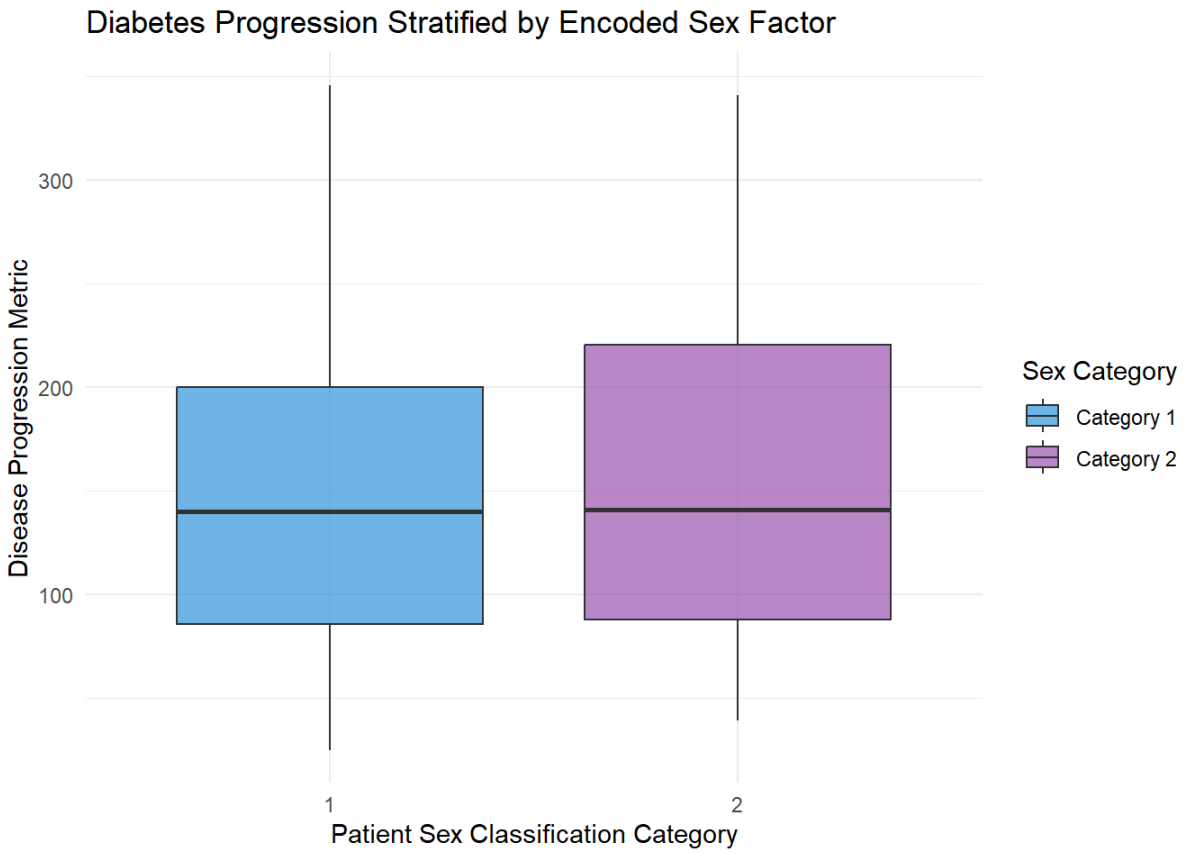



```
# Visualization 2: Histogram mapping baseline BMI metrics
ggplot(diabetes, aes(x = bmi)) +
  geom_histogram(bins = 25, fill = "#16a085", colour = "white", alpha = 0.8) +
  labs(
    title = "Distribution of Baseline Body Mass Index (BMI)",
    subtitle = "Clinical overweight threshold indicated at BMI > 25",
    x = "Body Mass Index (kg/m²)",
    y = "Patient Count (Frequency)"
  ) +
  geom_vline(xintercept = 25, linetype = "dashed", color = "#e74c3c", size = 1) +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
# Visualization 3 (Additional visual requirement): Boxplot of progression stratified by sex
ggplot(diabetes, aes(x = sex, y = progression, fill = sex)) +
  geom_boxplot(alpha = 0.7, outlier.colour = "#e74c3c") +
  scale_fill_manual(values = c("#3498db", "#9b59b6"), labels = c("Category 1", "Category 2")) +
  labs(
    title = "Diabetes Progression Stratified by Encoded Sex Factor",
    x = "Patient Sex Classification Category",
    y = "Disease Progression Metric",
    fill = "Sex Category"
  ) +
  theme_minimal()
```




```
# ========================================================================
# QUESTION 3. INVESTIGATE RELATIONSHIPS
# ========================================================================

# Isolate numeric continuous attributes only (excluding factor 'sex' from Pearson matrix)
numeric_data <- diabetes %>% select(where(is.numeric))

# Calculate and print the bivariate linear Pearson correlation matrix
cor_matrix <- cor(numeric_data, use = "complete.obs", method = "pearson")
print("--- Rounded Pearson Correlation Matrix ---")
```

```
## [1] "--- Rounded Pearson Correlation Matrix ---"
```

```
print(round(cor_matrix, 2))
```

```
##                age   bmi    bp   s1    s2    s3    s4    s5    s6 progression
## age           1.00  0.19  0.34 0.26  0.22 -0.08  0.20  0.27  0.30        0.19
## bmi           0.19  1.00  0.40 0.25  0.26 -0.37  0.41  0.45  0.39        0.59
## bp            0.34  0.40  1.00 0.24  0.19 -0.18  0.26  0.39  0.39        0.44
## s1            0.26  0.25  0.24 1.00  0.90  0.05  0.54  0.52  0.33        0.21
## s2            0.22  0.26  0.19 0.90  1.00 -0.20  0.66  0.32  0.29        0.17
## s3           -0.08 -0.37 -0.18 0.05 -0.20  1.00 -0.74 -0.40 -0.27       -0.39
## s4            0.20  0.41  0.26 0.54  0.66 -0.74  1.00  0.62  0.42        0.43
## s5            0.27  0.45  0.39 0.52  0.32 -0.40  0.62  1.00  0.46        0.57
## s6            0.30  0.39  0.39 0.33  0.29 -0.27  0.42  0.46  1.00        0.38
## progression   0.19  0.59  0.44 0.21  0.17 -0.39  0.43  0.57  0.38        1.00
```

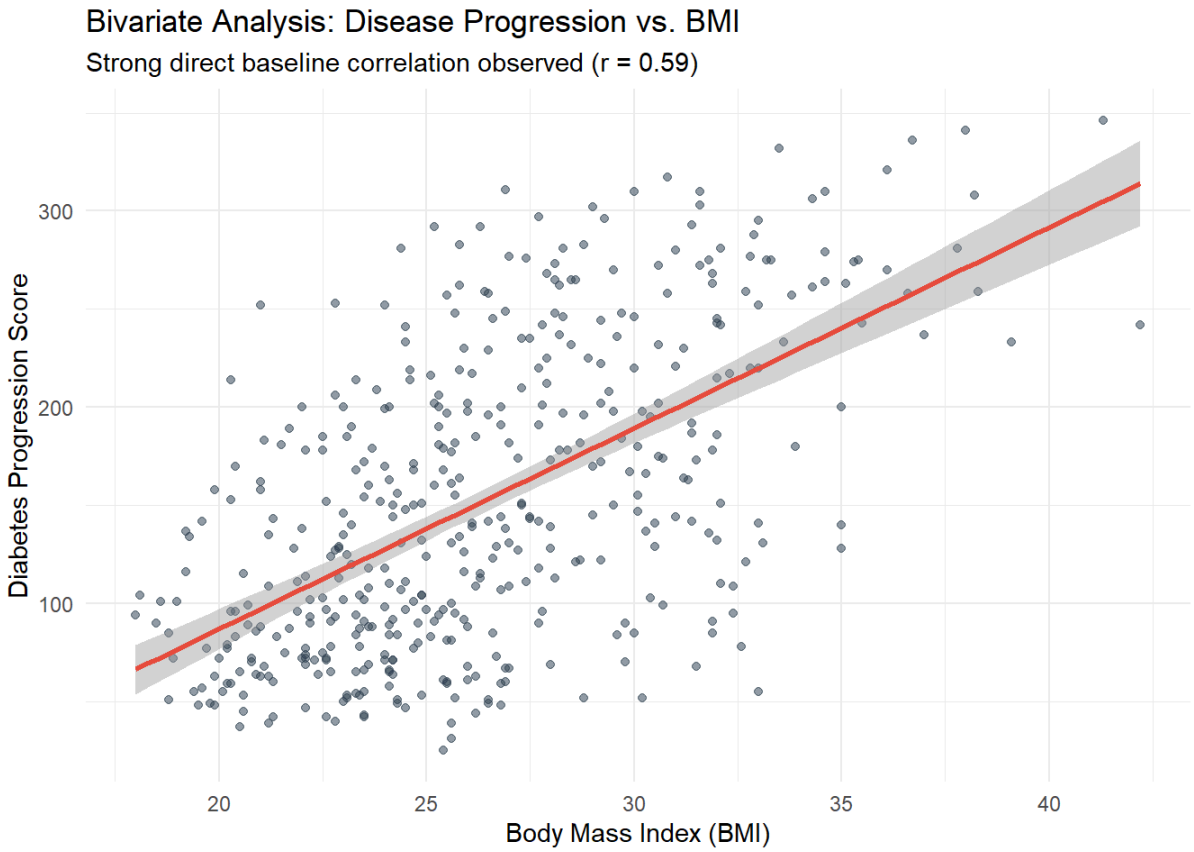
```
# Rank associations linked with the primary progression esito in descending order
progression_correlations <- sort(cor_matrix[, "progression"], decreasing = TRUE)
print("--- Ranked Direct Linear Correlations with Progression ---")
```

```
## [1] "--- Ranked Direct Linear Correlations with Progression ---"
```

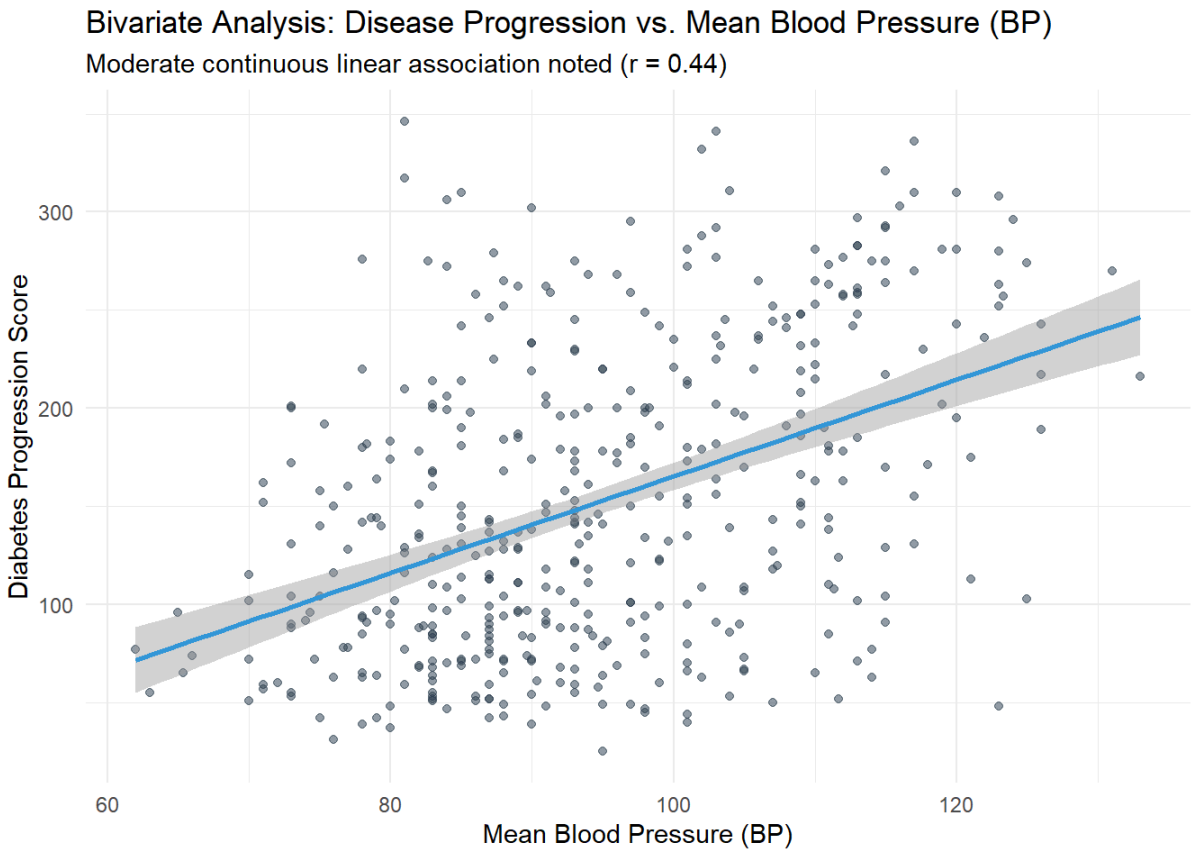
```
print(progression_correlations)
```

```
## progression         bmi          s5          bp          s4          s6
##      1.0000      0.5865      0.5659      0.4415      0.4305      0.3825
##          s1         age          s2          s3
##      0.2120      0.1879      0.1741     -0.3948
```

```
# Scatterplot 1: Linear relationship evaluation between BMI and Progression
ggplot(diabetes, aes(x = bmi, y = progression)) +
  geom_point(color = "#2c3e50", alpha = 0.5) +
  geom_smooth(method = "lm", formula = y ~ x, color = "#e74c3c", se = TRUE) +
  labs(
    title = "Bivariate Analysis: Disease Progression vs. BMI",
    subtitle = "Strong direct baseline correlation observed (r = 0.59)",
    x = "Body Mass Index (BMI)",
    y = "Diabetes Progression Score"
  ) +
  theme_minimal()
```




```
# Scatterplot 2: Linear relationship evaluation between Blood Pressure and Progression
ggplot(diabetes, aes(x = bp, y = progression)) +
  geom_point(color = "#2c3e50", alpha = 0.5) +
  geom_smooth(method = "lm", formula = y ~ x, color = "#3498db", se = TRUE) +
  labs(
    title = "Bivariate Analysis: Disease Progression vs. Mean Blood Pressure (BP)",
    subtitle = "Moderate continuous linear association noted (r = 0.44)",
    x = "Mean Blood Pressure (BP)",
    y = "Diabetes Progression Score"
  ) +
  theme_minimal()
```




```
# ================================================================================
# QUESTION 4. BUILD AND INTERPRET REGRESSION MODELS
# ================================================================================

# 1. Fit the Baseline Simple Linear Regression Model (Single Predictor: BMI)
simple_model <- lm(progression ~ bmi, data = diabetes)
print("--- SIMPLE LINEAR REGRESSION MODEL SUMMARY (SOLO BMI) ---")
```

```
## [1] "--- SIMPLE LINEAR REGRESSION MODEL SUMMARY (SOLO BMI) ---"
```

```
summary(simple_model)
```

```
##
## Call:
## lm(formula = progression ~ bmi, data = diabetes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -164.92  -43.57   -8.65   46.34  154.88
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -117.773     18.019   -6.54  1.8e-10 ***
## bmi           10.233      0.674   15.19  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 62.5 on 440 degrees of freedom
## Multiple R-squared:  0.344,  Adjusted R-squared:  0.342
## F-statistic:  231 on 1 and 440 DF,  p-value: <2e-16
```

```r
# 2. Fit the Optimized Structural Multiple Linear Regression Model (Three Pillars)
multiple_model <- lm(progression ~ bmi + bp + s5, data = diabetes)
print("--- OPTIMIZED MULTIPLE LINEAR REGRESSION MODEL SUMMARY ---")
```

```
## [1] "--- OPTIMIZED MULTIPLE LINEAR REGRESSION MODEL SUMMARY ---"
```

```r
summary(multiple_model)
```

```
##
## Call:
## lm(formula = progression ~ bmi + bp + s5, data = diabetes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -140.23  -40.64   -2.19   38.27  139.80
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -334.881     25.878  -12.94  < 2e-16 ***
## bmi            6.500      0.697    9.32  < 2e-16 ***
## bp             0.903      0.217    4.17  3.7e-05 ***
## s5            49.577      5.890    8.42  5.6e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 55.8 on 438 degrees of freedom
## Multiple R-squared:  0.48,   Adjusted R-squared:  0.477
## F-statistic:  135 on 3 and 438 DF,  p-value: <2e-16
```

```r
# Extract and bundle performance fit parameters (R-squared and Residual Error)
model_comparison <- bind_rows(
  glance(simple_model) %>% mutate(model = "Simple Linear: Solo BMI"),
  glance(multiple_model) %>% mutate(model = "Multiple Linear: BMI + BP + S5")
) %>%
  select(model, r.squared, adj.r.squared, sigma)

print("--- PREDICTIVE PERFORMANCE COMPONENT COMPARISON ---")
```

```
## [1] "--- PREDICTIVE PERFORMANCE COMPONENT COMPARISON ---"
```

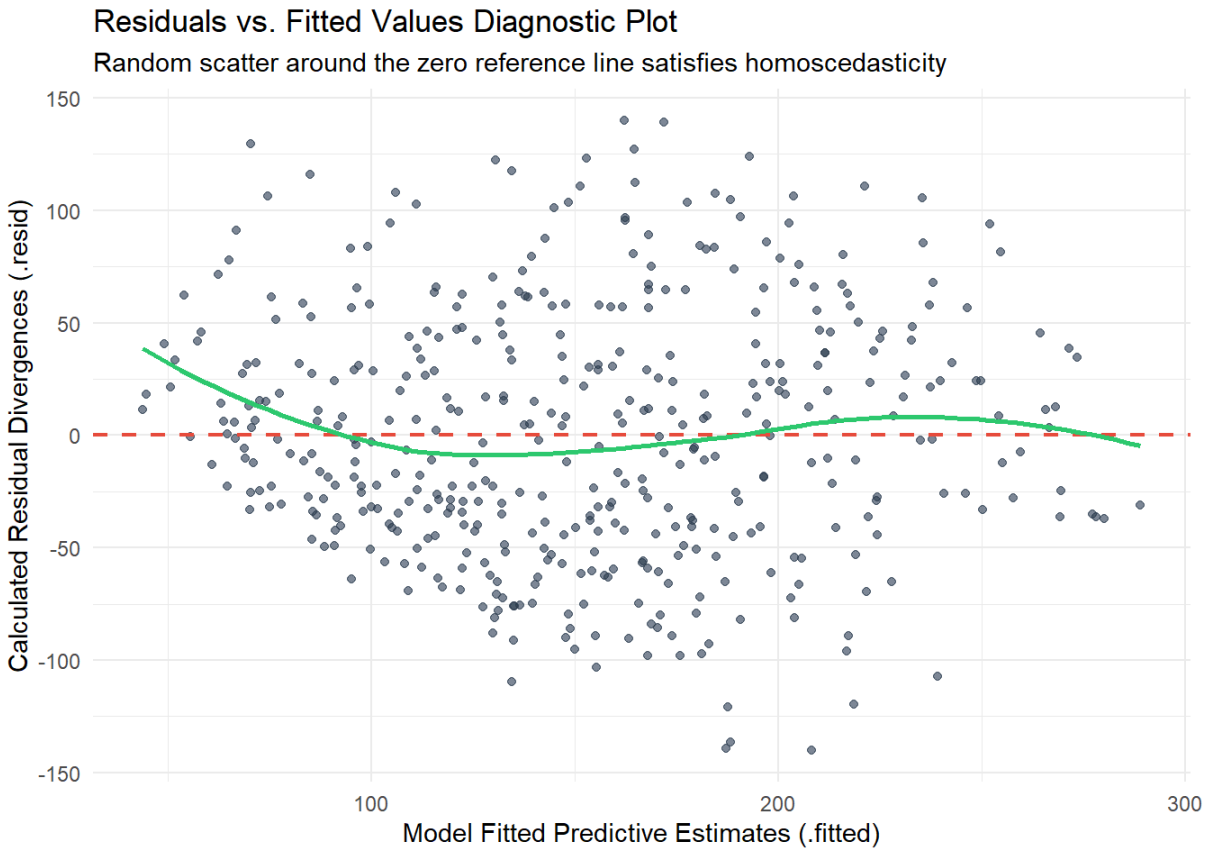
```r
print(model_comparison)
```

```
## # A tibble: 2 × 4
##   model                          r.squared adj.r.squared sigma
##   <chr>                              <dbl>         <dbl> <dbl>
## 1 Simple Linear: Solo BMI            0.344         0.342  62.5
## 2 Multiple Linear: BMI + BP + S5     0.480         0.477  55.8
```

```
# ==============================================================================
# QUESTION 5. EVALUATE THE ANALYSIS
# ==============================================================================

# Extract diagnostic mapping variables (Fitted estimates and Residual distances)
diagnostic_data <- augment(multiple_model)

# Residual Diagnostic Mapping: Residuals vs. Fitted Plot (Homoscedasticity Audit)
ggplot(diagnostic_data, aes(x = .fitted, y = .resid)) +
  geom_point(color = "#2c3e50", alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#e74c3c", size = 0.8) +
  geom_smooth(method = "loess", formula = y ~ x, color = "#2ecc71", se = FALSE) +
  labs(
    title = "Residuals vs. Fitted Values Diagnostic Plot",
    subtitle = "Random scatter around the zero reference line satisfies homoscedasticity",
    x = "Model Fitted Predictive Estimates (.fitted)",
    y = "Calculated Residual Divergences (.resid)"
  ) +
  theme_minimal()
```




```
# Code loop block to compute the Variance Inflation Factor (VIF) manually
manual_vif <- function(model) {
  X <- model.matrix(model)[, -1, drop = FALSE] # Remove intercept term
  vif_values <- sapply(seq_len(ncol(X)), function(i) {
    target <- X[, i]
    others <- X[, -i, drop = FALSE]
    if (ncol(others) == 0) return(1)
    r2 <- summary(lm(target ~ others))$r.squared
    1 / (1 - r2)
  })
  setNames(vif_values, colnames(X))
}

print("--- MULTICOLLINEARITY STRUCTURAL AUDIT (VIF) ---")
```

```
## [1] "--- MULTICOLLINEARITY STRUCTURAL AUDIT (VIF) ---"
```

```
print(manual_vif(multiple_model)) # VIF scores around 1.3 verify stable predictors
```

```
##   bmi    bp    s5
## 1.345 1.274 1.342
```

```
print("--- 95% STATISTICAL CONFIDENCE INTERVALS FOR COEFFICIENTS ---")
```

```
## [1] "--- 95% STATISTICAL CONFIDENCE INTERVALS FOR COEFFICIENTS ---"
```

```
print(confint(multiple_model, level = 0.95))
```

```
##                 2.5 %   97.5 %
## (Intercept) -385.7426 -284.020
## bmi            5.1300    7.870
## bp             0.4769    1.329
## s5            38.0002   61.154
```

```
# ================================================================================
# QUESTION 6. DRAW APPROPRIATE CONCLUSIONS
# ================================================================================
```